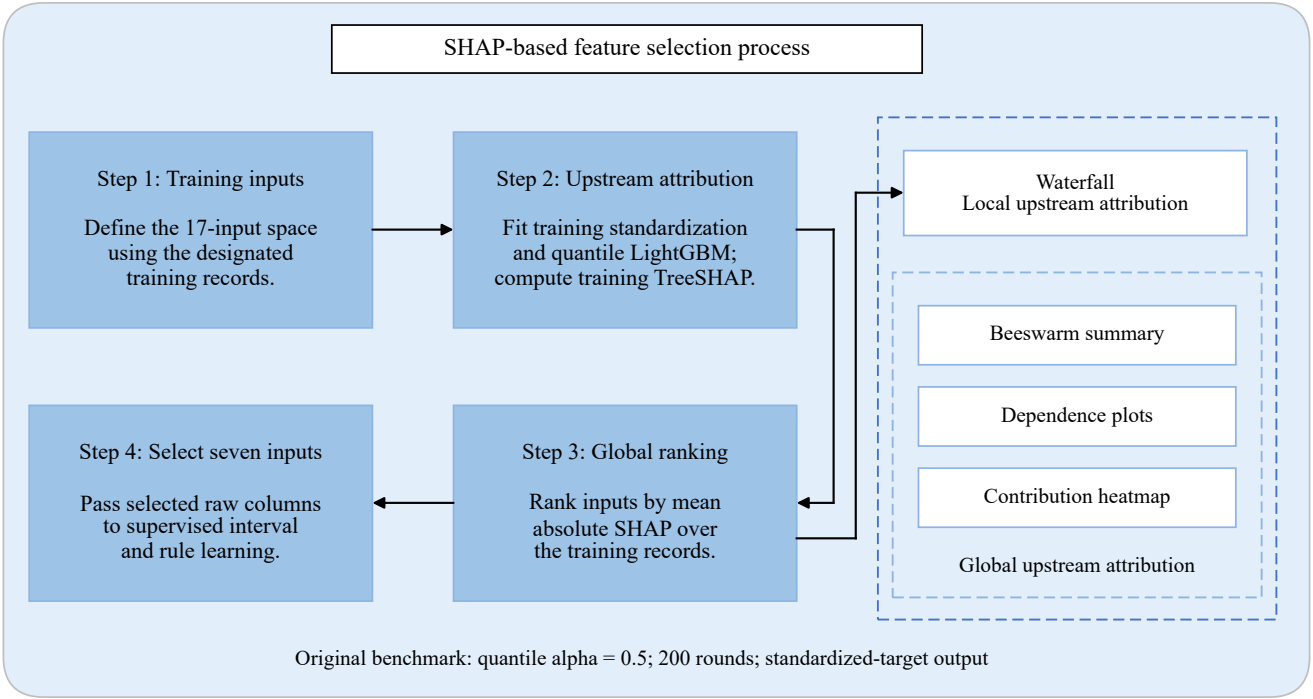
SHAP-based feature selection process
Step 1: Training inputs
Define the 17-input space using the designated training records.
Step 2: Upstream attribution
Fit training standardization and quantile LightGBM; compute training TreeSHAP.
Step 3: Global ranking
Rank inputs by mean absolute SHAP over the training records.
Step 4: Select seven inputs
Pass selected raw columns to supervised interval and rule learning.
Waterfall
Local upstream attribution
Beeswarm summary
Dependence plots
Contribution heatmap
Global upstream attribution
Original benchmark: quantile alpha = 0.5; 200 rounds; standardized-target output